1.58 Large Data Set: Edexcel

Worked Solutions

1 a)

Camborne, Heathrow, Hurn, Leeming and Leuchars

1 b)

Beijing, Jacksonville and Perth

1 c)

1987 and 2015

1 d)

May to October inclusive – which is 184 days

2 a)

Camborne, Hurn, Leuchars, Jacksonville and Perth

2 b)

Perth

3)

Perth is experiencing winter months due to being in the southern hemisphere

4)

Leuchars

5 a)

°C

5 b)

mm

5 c)

Hours

5 d)

None (it's a %)

5 e)

Knots or the Beaufort Scale

5 f)

Oktas (number out of 8)

5 g)

decametres

5 h)

hPa (hectopascals)

6 a)

3 knots $= 1.15 \times 3 = 3.45$ mph

6 b)

5.75mph $= 5.75 \div 1.15 = 5$ knots

6 c)

30mph $= 30 \div 1.15 = 26.1$ knots

7 a)

$\frac{1}{8}$ of the sky has could coverage

7 b)

$\frac{1}{2}$ of the sky has cloud coverage

7 c)

There is full cloud coverage

8)

Note that: The daily mean wind direction is rounded to the nearest 10 degrees. The original data point can be found in the Large Data Set.

WNW

9)

The cloud coverage is negatively skewed across the data set so it is more likely to be high cloud coverage.

10)

She is ignoring all the data points with the letters "tr" which represents a small amount of rain, missing 6 days which she should have put as 0 mm of rainfall.

By including those, the mean and the standard deviation will both go down.

11 a)

$$\bar{x} = \frac{321.2}{62} = 5.18 \text{ hours}$$

$$s_x = \sqrt{\frac{2692.66 - \frac{(321.2)^2}{62}}{61}} = 4.11$$

11 b)

The average amount of hours of daily sunshine has gone down since 1987 but the spread of the hours has roughly stayed the same (there has been a slight decrease in spread).

12 a)

September & October due to the coldest days

12 b)

Invalid claim as it is only a comparison of two months and only in one location.

13 a)

Systematic sampling

13 b)

He didn't choose a random start point, between 1 and 6, he started with the first date.

14 a)

$$10k + 0.04 + 0.09 + 0.14 + 0.17 + 0.22 + 0.23 + k = 1$$

$$11k + 0.89 = 1 \Rightarrow k = 0.01$$

$\mathrm{P}(c = 1) = 0.01$ and $\mathrm{P}(c = 8) = 0.1$

14 b)

$$\mathrm{P}(c \leq 3) = 0.01 + 0.04 + 0.09 = 0.14$$

$0.14 \times 184 = 25.76$ so 26 days

14 c)

$\mathrm{P}(X \leq 3) = 0.1358$ so $24.98927\ldots = 25$ days

14 d)

For this distribution there are more than the 0 to 8 oktas, $\mathrm{P}(X \geq 9) = 1 - 0.8597 = 0.1403$ meaning that around 26 days are 9 or more Oktas which doesn't exist!

15 a)

$0.151 \times 1026 - 148 = 6.926$ knots

15 b)

There is a high correlation coefficient, and whilst there should be a positive correlation, the magnitude of PMCC value of this data set is too strong.

16 a)

freq	x	fx	fx^2
0.076	4	0.304	1.216
0.272	6	1.632	9.792
0.321	8	2.568	20.544
0.206	10	2.06	20.6
0.076	12	0.912	10.944
0.049	14	0.686	9.604
1		8.162	72.7

Mean is 8.162

Std dev: $\sqrt{\frac{72.7}{1} - (8.162)^2} = 2.47$ (3 SF)

16 b)

$$1 \times 0.076 + 2 \times 0.049 = 0.174$$

Total area $= 0.152 + 0.544 + 0.642 + 0.412 + 0.152 + 0.098 = 2$

Area of 2 for 184 days (number of days from knowledge of Large Data Set)

$0.174 \times 92 = 16.008$ so 16 days where the daily mean windspeed is over 12 knots

16 c)

$$X \sim \mathrm{N}(8.162, 2.47^2)$$

$$\mathrm{P}(X > 12) = 0.0601$$

16 d)

For $X \sim \mathrm{N}(8.162, 2.47^2)$

$$2.47 \times 3 = 7.41$$

So distribution lies approximately between 0.752 and 15.572 which is reasonable.

The mean is 8.162 and the median is ≈ 7.95, and the mode ≈ 8 which are similar to each other, this indicates the symmetry needed for a normal distribution.

17 a)

Let ρ be the population correlation coefficient between the windspeed in Leeming on a given day and the relative humidity in Leeming on that same day.

$\mathrm{H}_0 : \rho = 0$
$\mathrm{H}_1 : \rho < 0$
$r_s = -0.3613$

The critical value for $n = 20$ for a one-tailed test at a 5% significance level is 0.3783

$-0.3613 > -0.3783$ so we fail to reject H_0

There is insufficient evidence to suggest that there is a negative correlation between the windspeed in Leeming on a given day and the relative humidity in Leeming on that same day.

17 b)

$-0.6234 < -0.3783$ so we reject H_0

There is sufficient evidence to suggest that there is a negative correlation between the windspeed in Leeming on a given day and the relative humidity in Leeming on that same day.